UMA033: NUMERICAL AND STATISTICAL METHODS				
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	3	0	2	4

Course Objective:The main objective of this course is to understand and implement various numerical and statistical methods to solve engineering, physical and real life problems.

Syllabus

Basic of Errors: Floating-point representation, rounding and chopping errors.
Non-Linear Equations: Bisection, fixed point iteration, Newton – Raphson's method for simple and multiple roots and order of convergence.
Linear Systems and Eigen-Values: Gauss elimination method using partial pivoting, Gauss--Seidel method, Rayleigh's power method for eigen-values and eigen-vectors.
Interpolation and Approximations: Newton's forward and backward differences, Lagrange (with error analysis), Newton's divided difference interpolation formulas.
Numerical Integration: Newton-Cotes quadrature formulae (Trapezoidal and Simpson's rules) and their error analysis, Gauss-Legendre quadrature formulae.
Differential Equations: Solution of initial value problems using Euler's, Modified Euler's and Runge-Kutta methods (fourth-order).
Curve Fitting: Curve fitting by the method of least squares- fitting of straight lines, second degree parabolas and more general curves.
Probability Distribution: Mathematical expectations, Definition of probability distribution (Probability Mass Function and Probability Density Function), Poisson, Geometric, Binomial, Uniform and Normal distributions.
Correlation and Regression: Bivariate distribution, correlation coefficients, regression lines, formula for regression coefficients.

Laboratory Work: Lab experiments will be set in consonance with materials covered in the theory using MATLAB.

Course Learning Objectives (CLO)

The students will be able to:

1. Learn how to obtain numerical solution of nonlinear equations using bisection, Newton and fixed-point iteration methods.
2. Solve system of linear equations numerically using direct and iterative methods.
3. Analyze the correlated data using the least square and regression curves.
4. Solve integration and initial value problems numerically.
5. Solve real life problems using various probability distributions.
6. Approximate the data and functions using interpolating polynomials.

Text Books

1. *K. Atkinson and W. Han, Elementary Numerical Analysis, 3rd edition, John Wiley & Sons, 2004.*
2. *Brian Bradie, A friendly Introduction to Numerical Analysis, prentice Hall, 2007.*
3. *Burden L. R., Faires D. J. and Burden A.M., Numerical Analysis, Brooks Cole, 8th edition, 2004.*
4. *Richards A. Johnson, Probability and Statistics for Engineers, 8th Edition, PHI Learning, 2011.*
5. *Meyer, P.L., Introductory Probability and Statistical applications, 2nd edition, Oxford, 1970*

Reference Books

1. *Curtis, F. Gerald and Patrick O. Wheatley, Applied Numerical Analysis, 7th edition Pearson Education, 2003.*
2. *Walpole, Ronald E., Myers, Raymond H. Myers, and Sharon L. Myers, Probability and Statistics forEngineers and Scientists, 8th edition Pearson Education, 2007.*
3. *Steven C. Chapra, Applied Numerical Methods with MATLAB for Engineers and Scientists, McGraw-Hill Publishing; 2nd edition, 2007.*